

1 Composite Numerical Integration

The Newton-Cotes formulas are generally unsuitable for use over large integration intervals. High-degree formulas would be required, and the values of the coefficients in these formulas are difficult to obtain. Also, the Newton-Cotes formulas are based on interpolatory polynomials that use equally-spaced nodes, a procedure that is inaccurate over large intervals because of the oscillatory nature of high-degree polynomials.

In this section, we discuss a piecewise approach to numerical integration that uses the low-order Newton-Cotes formulas. These are the techniques most often applied.

Example 1. Use Simpson's rule to approximate $\int_0^4 e^x dx$ and compare this to the results obtained by adding the Simpson's rule approximations for $\int_0^2 e^x dx$ and $\int_2^4 e^x dx$. Compare these approximations to the sum of Simpson's rule for $\int_0^1 e^x dx$, $\int_1^2 e^x dx$, $\int_2^3 e^x dx$, and $\int_3^4 e^x dx$.

Solution: Simpson's rule on $[0, 4]$ uses $h = 2$ and gives

$$\int_0^4 e^x dx \approx \frac{2}{3} (e^0 + 4e^2 + e^4) = 56.76958$$

The exact answer in this case is $e^4 - e^0 = 53.59815$, and the error -3.17143 is far larger than we would normally accept.

Applying Simpson's rule on each of the intervals $[0, 2]$ and $[2, 4]$ uses $h = 1$ and gives

$$\begin{aligned} \int_0^4 e^x dx &= \int_0^2 e^x dx + \int_2^4 e^x dx \\ &\approx \frac{1}{3} (e^0 + 4e + e^2) + \frac{1}{3} (e^2 + 4e^3 + e^4) \\ &= \frac{1}{3} (e^0 + 4e + 2e^2 + 4e^3 + e^4) \\ &= 53.86385 \end{aligned}$$

The error has been reduced to -0.26570 .

For the integrals on $[0, 1]$, $[1, 2]$, $[2, 3]$, and $[3, 4]$ we use Simpson's rule four times with $h = \frac{1}{2}$ giving

$$\begin{aligned}
\int_0^4 e^x dx &= \int_0^1 e^x dx + \int_1^2 e^x dx + \int_2^3 e^x dx + \int_3^4 e^x dx \\
&\approx \frac{1}{6} (e_0 + 4e^{1/2} + e) + \frac{1}{6} (e + 4e^{3/2} + e^2) \\
&\quad + \frac{1}{6} (e^2 + 4e^{5/2} + e^3) + \frac{1}{6} (e^3 + 4e^{7/2} + e^4) \\
&= \frac{1}{6} (e^0 + 4e^{1/2} + 2e + 4e^{3/2} + 2e^2 + 4e^{5/2} + 2e^3 + 4e^{7/2} + e^4) \\
&= 53.61622
\end{aligned}$$

The error for this approximation has been reduced to -0.01807 .

To generalize this procedure for an arbitrary integral $\int_a^b f(x)dx$, choose an even integer n . Subdivide the interval $[a, b]$ into n subintervals, and apply Simpson's rule on each consecutive pair of subintervals. (See Figure 1)

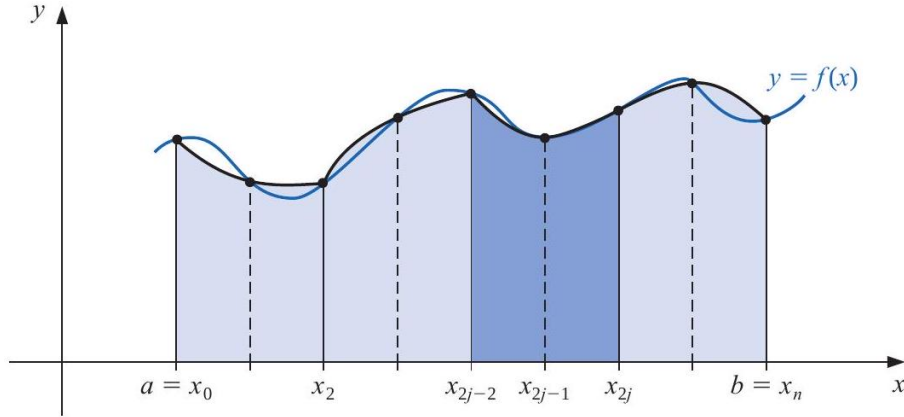


Figure 1: Figure 1

With $h = (b - a)/n$ and $x_j = a + jh$, for each $j = 0, 1, \dots, n$, we have

$$\begin{aligned}
\int_a^b f(x)dx &= \sum_{j=1}^{n/2} \int_{x_{2j-2}}^{x_{2j}} f(x)dx \\
&= \sum_{j=1}^{n/2} \left\{ \frac{h}{3} [f(x_{2j-2}) + 4f(x_{2j-1}) + f(x_{2j})] - \frac{h^5}{90} f^{(4)}(\xi_j) \right\}
\end{aligned}$$

for some ξ_j with $x_{2j-2} < \xi_j < x_{2j}$, provided that $f \in C^4[a, b]$. Using the fact that for each $j = 1, 2, \dots, (n/2) - 1$ we have $f(x_{2j})$ appearing in the term

corresponding to the interval $[x_{2j-2}, x_{2j}]$ and also in the term corresponding to the interval $[x_{2j}, x_{2j+2}]$, we can reduce this sum to

$$\int_a^b f(x)dx = \frac{h}{3} \left[f(x_0) + 2 \sum_{j=1}^{(n/2)-1} f(x_{2j}) + 4 \sum_{j=1}^{n/2} f(x_{2j-1}) + f(x_n) \right] - \frac{h^5}{90} \sum_{j=1}^{n/2} f^{(4)}(\xi_j)$$

The error associated with this approximation is

$$E(f) = -\frac{h^5}{90} \sum_{j=1}^{n/2} f^{(4)}(\xi_j)$$

where $x_{2j-2} < \xi_j < x_{2j}$, for each $j = 1, 2, \dots, n/2$.

These observations produce the following result.

Theorem. Let $f \in C^4[a, b]$, n be even, $h = (b - a)/n$, and $x_j = a + jh$, for each $j = 0, 1, \dots, n$. There exists a $\mu \in (a, b)$ for which the Composite Simpson's rule for n subintervals can be written with its error term as

$$\int_a^b f(x)dx = \frac{h}{3} \left[f(a) + 2 \sum_{j=1}^{(n/2)-1} f(x_{2j}) + 4 \sum_{j=1}^{n/2} f(x_{2j-1}) + f(b) \right] - \frac{b-a}{180} h^4 f^{(4)}(\mu)$$

Notice that the error term for the Composite Simpson's rule is $O(h^4)$, whereas it was $O(h^5)$ for the standard Simpson's rule. However, these rates are not comparable because for standard Simpson's rule we have h fixed at $h = (b - a)/2$, but for Composite Simpson's rule we have $h = (b - a)/n$, for n an even integer. This permits us to considerably reduce the value of h when the Composite Simpson's rule is used.

The subdivision approach can be applied to any of the Newton-Cotes formulas. The extensions of the Trapezoidal (see Figure 4.8) and Midpoint rules are given without proof. The Trapezoidal rule requires only one interval for each application, so the integer n can be either odd or even.

Theorem. Let $f \in C^2[a, b]$, $h = (b - a)/n$, and $x_j = a + jh$, for each $j = 0, 1, \dots, n$. There exists a $\mu \in (a, b)$ for which the **Composite Trapezoidal rule** for n subintervals can be written with its error term as

$$\int_a^b f(x)dx = \frac{h}{2} \left[f(a) + 2 \sum_{j=1}^{n-1} f(x_j) + f(b) \right] - \frac{b-a}{12} h^2 f''(\mu)$$

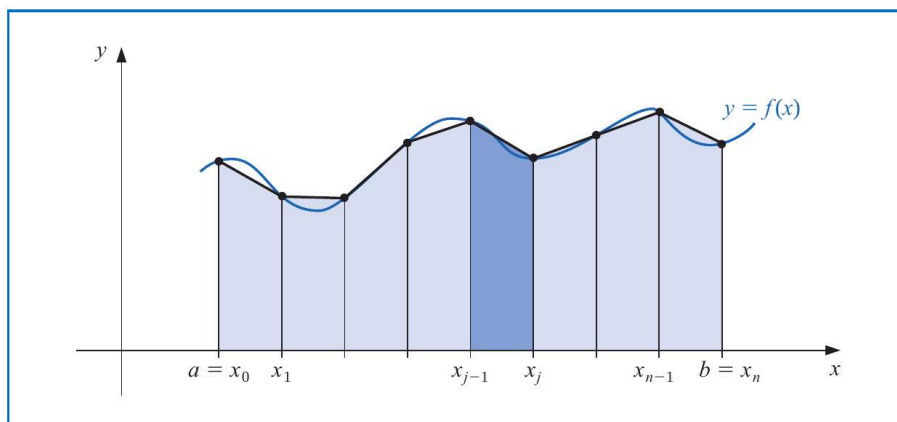


Figure 2: Figure 2

For the Composite Midpoint rule, n must again be even. (See Figure 3.)

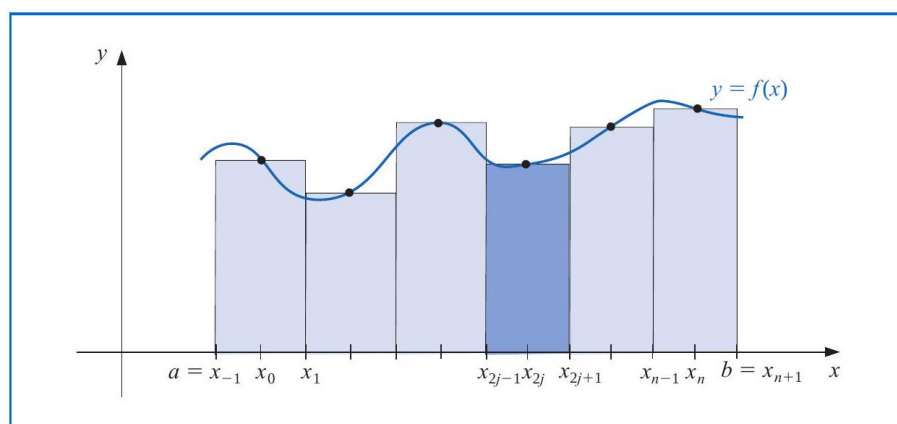


Figure 3: Figure 3

Theorem. Let $f \in C^2[a, b]$, n be even, $h = (b - a)/(n + 2)$, and $x_j = a + (j + 1)h$ for each $j = -1, 0, \dots, n + 1$. There exists a $\mu \in (a, b)$ for which the Composite Midpoint rule for $n + 2$ subintervals can be written with its error term as

$$\int_a^b f(x) dx = 2h \sum_{j=0}^{n/2} f(x_{2j}) + \frac{b-a}{6} h^2 f''(\mu)$$

Example 2. Determine values of h that will ensure an approximation error of less than 0.00002 when approximating $\int_0^\pi \sin x dx$ and employing (a) Composite Trapezoidal rule and (b) Composite Simpson's rule.

Solution:

- (a) The error form for the Composite Trapezoidal rule for $f(x) = \sin x$ on $[0, \pi]$ is

$$\left| \frac{\pi h^2}{12} f''(\mu) \right| = \left| \frac{\pi h^2}{12} (-\sin \mu) \right| = \frac{\pi h^2}{12} |\sin \mu|.$$

To ensure sufficient accuracy with this technique we need to have

$$\frac{\pi h^2}{12} |\sin \mu| \leq \frac{\pi h^2}{12} < 0.00002$$

Since $h = \pi/n$ implies that $n = \pi/h$, we need

$$\frac{\pi^3}{12n^2} < 0.00002 \quad \text{which implies that } n > \left(\frac{\pi^3}{12(0.00002)} \right)^{1/2} \approx 359.44$$

and the Composite Trapezoidal rule requires $n \geq 360$.

- (b) The error form for the Composite Simpson's rule for $f(x) = \sin x$ on $[0, \pi]$ is

$$\left| \frac{\pi h^4}{180} f^{(4)}(\mu) \right| = \left| \frac{\pi h^4}{180} \sin \mu \right| = \frac{\pi h^4}{180} |\sin \mu|.$$

To ensure sufficient accuracy with this technique we need to have

$$\frac{\pi h^4}{180} |\sin \mu| \leq \frac{\pi h^4}{180} < 0.00002$$

Using again the fact that $n = \pi/h$ gives

$$\frac{\pi^5}{180n^4} < 0.00002 \quad \text{which implies that } n > \left(\frac{\pi^5}{180(0.00002)} \right)^{1/4} \approx 17.07$$

So Composite Simpson's rule requires only $n \geq 18$.

Composite Simpson's rule with $n = 18$ gives

$$\int_0^\pi \sin x dx \approx \frac{\pi}{54} \left[2 \sum_{j=1}^8 \sin \left(\frac{j\pi}{9} \right) + 4 \sum_{j=1}^9 \sin \left(\frac{(2j-1)\pi}{18} \right) \right] = 2.0000104$$

This is accurate to within about 10^{-5} because the true value is $-\cos(\pi) - (-\cos(0)) = 2$.

Composite Simpson's rule is the clear choice if you wish to minimize computation. For comparison purposes, consider the Composite Trapezoidal rule using $h = \pi/18$ for the integral in Example 2. This approximation uses the same function evaluations as Composite Simpson's rule but the approximation in this case

$$\int_0^\pi \sin x dx \approx \frac{\pi}{36} \left[2 \sum_{j=1}^{17} \sin\left(\frac{j\pi}{18}\right) + \sin 0 + \sin \pi \right] = \frac{\pi}{36} \left[2 \sum_{j=1}^{17} \sin\left(\frac{j\pi}{18}\right) \right] = 1.9949205.$$

is accurate only to about 5×10^{-3} .

Round-Off Error Stability

In Example 2 we saw that ensuring an accuracy of 2×10^{-5} for approximating $\int_0^\pi \sin x dx$ required 360 subdivisions of $[0, \pi]$ for the Composite Trapezoidal rule and only 18 for Composite Simpson's rule. In addition to the fact that less computation is needed for the Simpson's technique, you might suspect that because of fewer computations this method would also involve less round-off error. However, an important property shared by all the composite integration techniques is a stability with respect to round-off error. That is, the round-off error does not depend on the number of calculations performed.

To demonstrate this rather amazing fact, suppose we apply the Composite Simpson's rule with n subintervals to a function f on $[a, b]$ and determine the maximum bound for the round-off error. Assume that $f(x_i)$ is approximated by $\tilde{f}(x_i)$ and that

$$f(x_i) = \tilde{f}(x_i) + e_i, \quad \text{for each } i = 0, 1, \dots, n,$$

where e_i denotes the round-off error associated with using $\tilde{f}(x_i)$ to approximate $f(x_i)$. Then the accumulated error, $e(h)$, in the Composite Simpson's rule is

$$\begin{aligned} e(h) &= \left| \frac{h}{3} \left[e_0 + 2 \sum_{j=1}^{(n/2)-1} e_{2j} + 4 \sum_{j=1}^{n/2} e_{2j-1} + e_n \right] \right| \\ &\leq \frac{h}{3} \left[|e_0| + 2 \sum_{j=1}^{(n/2)-1} |e_{2j}| + 4 \sum_{j=1}^{n/2} |e_{2j-1}| + |e_n| \right]. \end{aligned}$$

If the round-off errors are uniformly bounded by ε , then

$$e(h) \leq \frac{h}{3} \left[\varepsilon + 2 \left(\frac{n}{2} - 1 \right) \varepsilon + 4 \left(\frac{n}{2} \right) \varepsilon + \varepsilon \right] = \frac{h}{3} 3n\varepsilon = nh\varepsilon.$$

But $nh = b - a$, so

$$e(h) \leq (b - a)\varepsilon$$

a bound independent of h (and n). This means that, even though we may need to divide an interval into more parts to ensure accuracy, the increased computation that is required does not increase the round-off error. This result implies that the procedure is stable as h approaches zero. Recall that this was not true of the numerical differentiation procedures we have considered.